Basic Principles of Medicine 1

Module: Foundations to Medicine| Lecture 29

Connective Tissue Histology

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Required Reading

Objective: SOM.MK.I.BPM1.1.FTM.3.HCB.0856

Distinguish between the wandering cell types found in connective tissue (lymphocyte, plasma cell, neutrophil, monocyte, basophil and eosinophil) based on their location and functions.

Recommended Reading

Histology – A Text and Atlas

Pawlina 8th Edition

Chapter 6: Connective Tissue Pg. 176-215

https://periodicals.sgu.edu/login?url=https://meded-lwwhealthlibrary-com.periodicals.sgu.edu/book.aspx?bookid=2583

Your Objectives will show here!

Objectives

SOM.MK.I.BPM1.1.FTM.3.HCB.0853	List the function(s) of the different types of connective tissue
SOM.MK.I.BPM1.1.FTM.3.HCB.0854	Identify the major components of connective tissue
SOM.MK.I.BPM1.1.FTM.3.HCB.0855	Distinguish between the wandering cell types found in connective tissue (lymphocyte, plasma cell, neutrophil, monocyte, basophil and eosinophil) based on their location and functions.
SOM.MK.I.BPM1.1.FTM.3.HCB.0856	Distinguish between the resident cell types found in connective tissue (fibroblasts, myofibroblasts, macrophages, adipocytes, stem cells and mast cells) based on their microanatomical features and functions.
SOM.MK.I.BPM1.1.FTM.3.HCB.0857	Describe the role of connective tissue cells in anaphylactic shock.
SOM.MK.I.BPM1.1.FTM.3.HCB.0858	Describe the two components that make up the extracellular matrix of connective tissue.
SOM.MK.I.BPM1.1.FTM.3.HCB.0859	Define ground substance.
SOM.MK.I.BPM1.1.FTM.3.HCB.0860	Describe the structure of the three main components of the ground substance (proteoglycans, glycosaminoglycans, multiadhesive glycoproteins) with at least 2 examples of each main component.
SOM.MK.I.BPM1.1.FTM.3.HCB.0861	Give at least 2 examples of each main component of the ground substance.
SOM.MK.I.BPM1.1.FTM.3.HCB.0862	Describe the cell type responsible for the production of ground substance.
SOM.MK.I.BPM1.1.FTM.3.HCB.0863	List the cell types responsible for the production of collagen, elastic and reticular fibers.
SOM.MK.I.BPM1.1.FTM.3.HCB.0864	Distinguish between collagen, elastic and reticular fibers based on their basic structure and their function(s).
SOM.MK.I.BPM1.1.FTM.3.HCB.0865	Identify the histological staining of collagen, elastic & reticular fibers.
SOM.MK.I.BPM1.1.FTM.3.HCB.0866	List common locations of types I, II and III collagen and elastic fibers.
SOM.MK.I.BPM1.1.FTM.3.HCB.0867	Define “argyrophillic” and state the fiber of connective tissue that exhibit this property.
SOM.MK.I.BPM1.1.FTM.3.HCB.0868	List the common types of fibrillar collagens.
SOM.MK.I.BPM1.1.FTM.3.HCB.0869	List the common types basement membrane collagens.
SOM.MK.I.BPM1.1.FTM.3.HCB.0870	Describe the alterations in collagen synthesis that results in abnormal scar formation.
SOM.MK.I.BPM1.1.FTM.3.HCB.0871	List the three major classifications of connective tissue with their subclassification.
SOM.MK.I.BPM1.1.FTM.3.HCB.0872	Describe the two subtypes of embryonic connective tissue.

Objectives

SOM.MK.I.BPM1.1.FTM.3.HCB.0873	Describe the two general subtypes of connective tissue proper.
SOM.MK.I.BPM1.1.FTM.3.HCB.0874	Describe the structure and location of lamina propria
SOM.MK.I.BPM1.1.FTM.3.HCB.0875	List the types of specialized connective tissue.
SOM.MK.I.BPM1.1.FTM.3.HCB.0876	List common location sites for each type of connective tissue.
SOM.MK.I.BPM1.1.FTM.3.HCB.0877	Compare and contrast white and brown adipose tissue in relation to its appearance, location and function.
SOM.MK.I.BPM1.1.FTM.3.HCB.0878	List at least two commonly used stains to identify connective tissue under light microscopy
SOM.MK.I.BPM1.1.FTM.3.HCB.0879	Describe the factors that lead to formation of interstitial fluid, the formation of lymph, and the development of edema.

Overview

- **General structure and functions**
- **Components:**
    - Cells
    - Protein fibers
    - Ground substance
- **Classification**
- **Clinical Correlations**

Connective tissue

- The tissue that supports, binds, or separates more specialized tissues and organs
- Forms a **vast** and **continuous compartment** throughout the body
- Bounded by the **basal laminae** of the various epithelia and by the basal or external laminae of muscle cells and nerve-supporting cells.
- **Connective tissue** consists of
  1. **Cells**
  2. **Extracellular matrix (ECM)**:
     a) **Protein fibers:**
        1. Collagen
        2. Elastic
        3. Reticular

     b) **Ground substance:**
        1. Proteoglycans
        2. Multi-adhesive glycoproteins
        3. Glycosaminoglycans [GAGs]

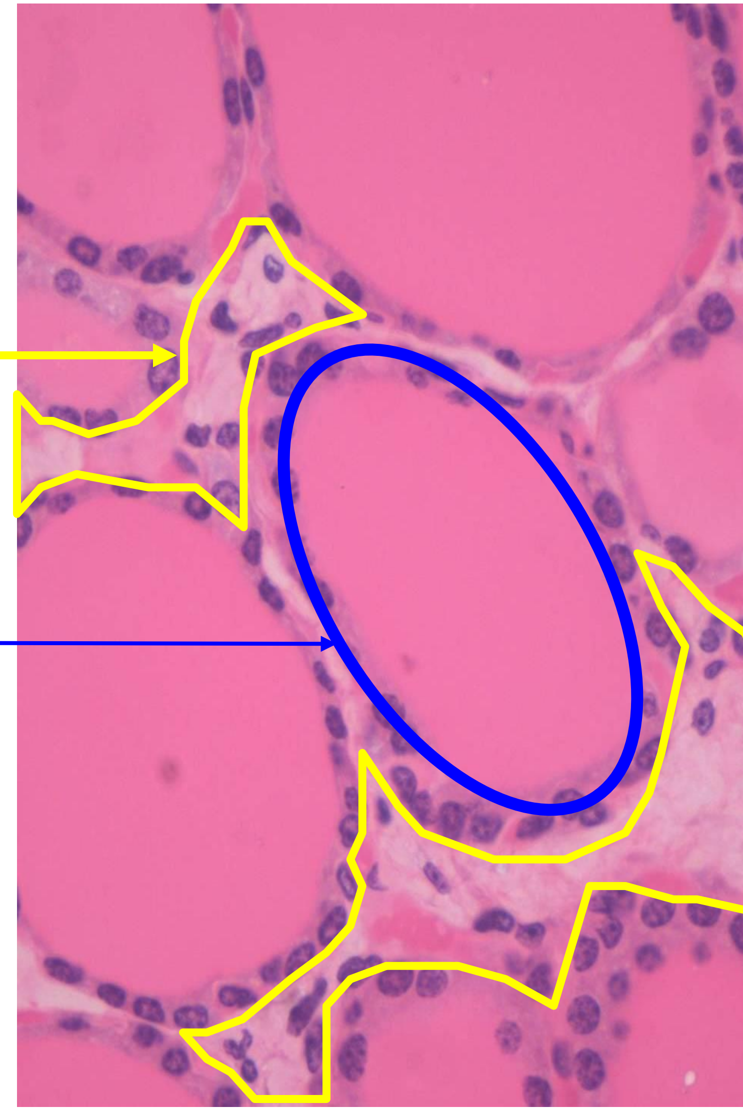
Functions of Connective Tissue

- Support
  - Forms the stroma of organs
- Binds tissue
- Repair

Stroma
(connective tissue)
• Supporting structures

VS

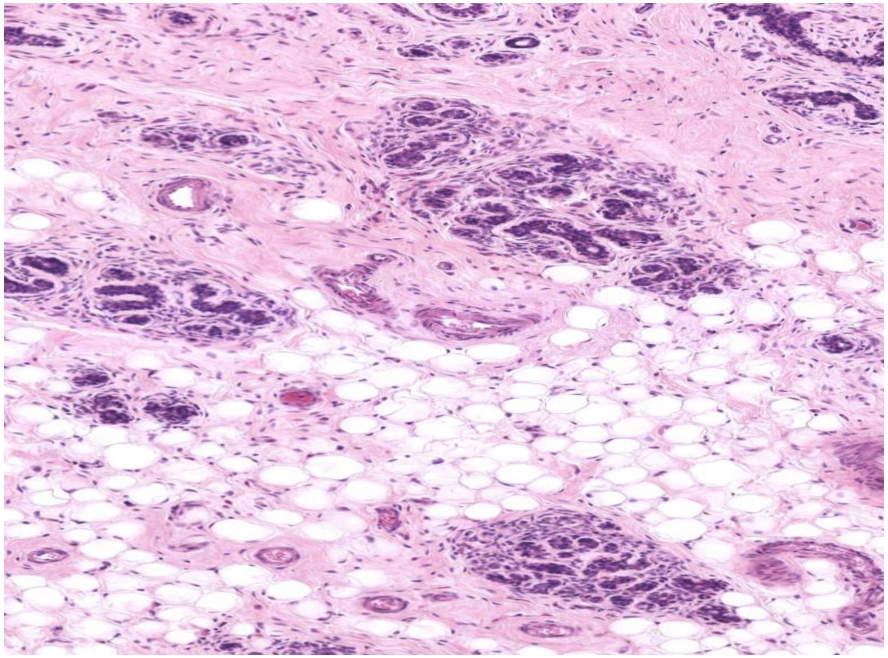
Parenchyma
(epithelium, nerve, muscle)
• Functional unit

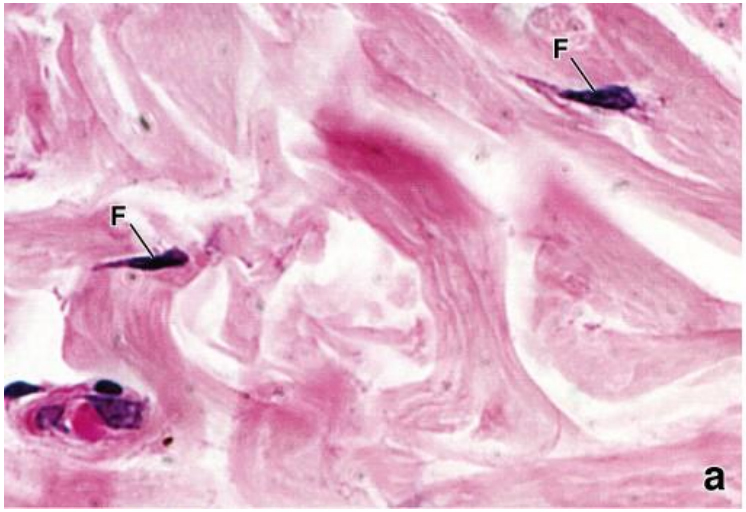


- Defense (immune function)
- Nutrition (storage & transport)
- Insulation
- Protection

Components of Connective Tissue

- **Cells**
  - Transient & permanent
- **Extracellular Matrix- ECM**
  - Ground substance
  - Fibers
- **General Features**:
- Highly vascularized and innervated
  - Exception: cartilage (avascular connective tissue)
- Directly supplied by lymphatic vessels
- All derived from mesenchyme





Connective Tissue: Cells

❖Resident/permanent

- Fibroblasts
- **Myofibroblasts**
- Macrophages
- Adipocytes
- Stem cells
- Mast cells

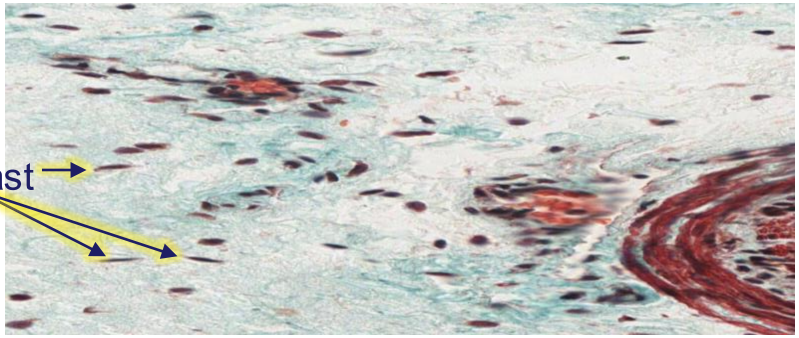
❖Wandering/transient

- consists primarily of cells that have migrated into the tissue from the blood in response to specific stimuli. These include:
  - Lymphocyte
  - Plasma cell
  - Neutrophil
  - Monocyte
  - Basophil
  - Eosinophil

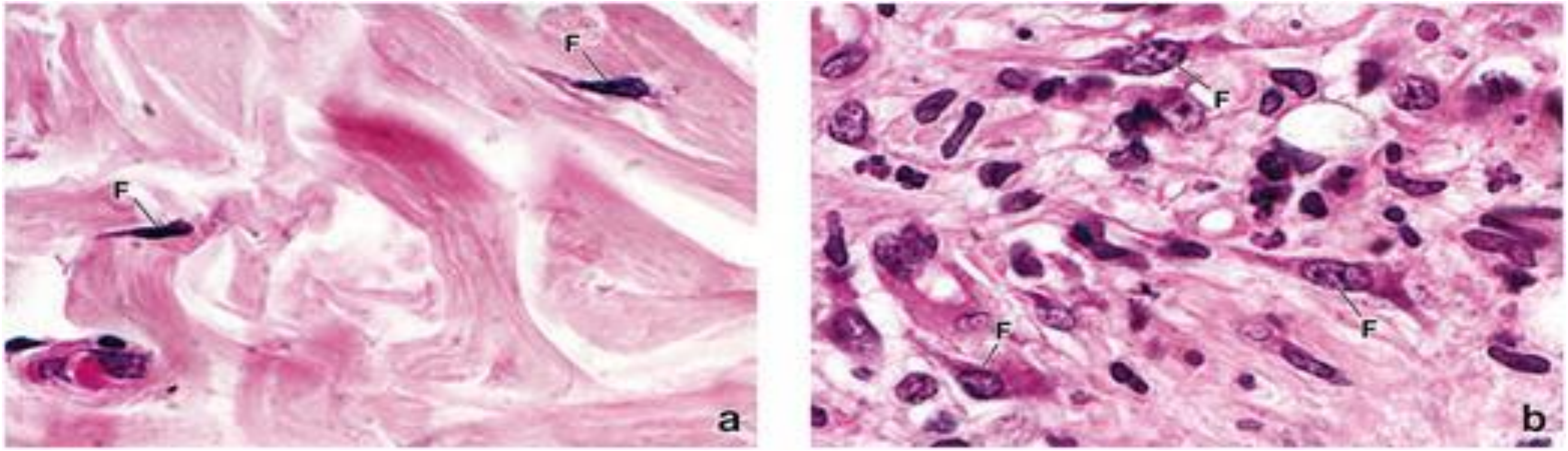
Fibroblasts

- Most common cell in connective tissue
- It resides near collagen fibers
- Synthesizes all the extracellular fibers
- Synthesizes and maintains the ground substance

Fibroblast

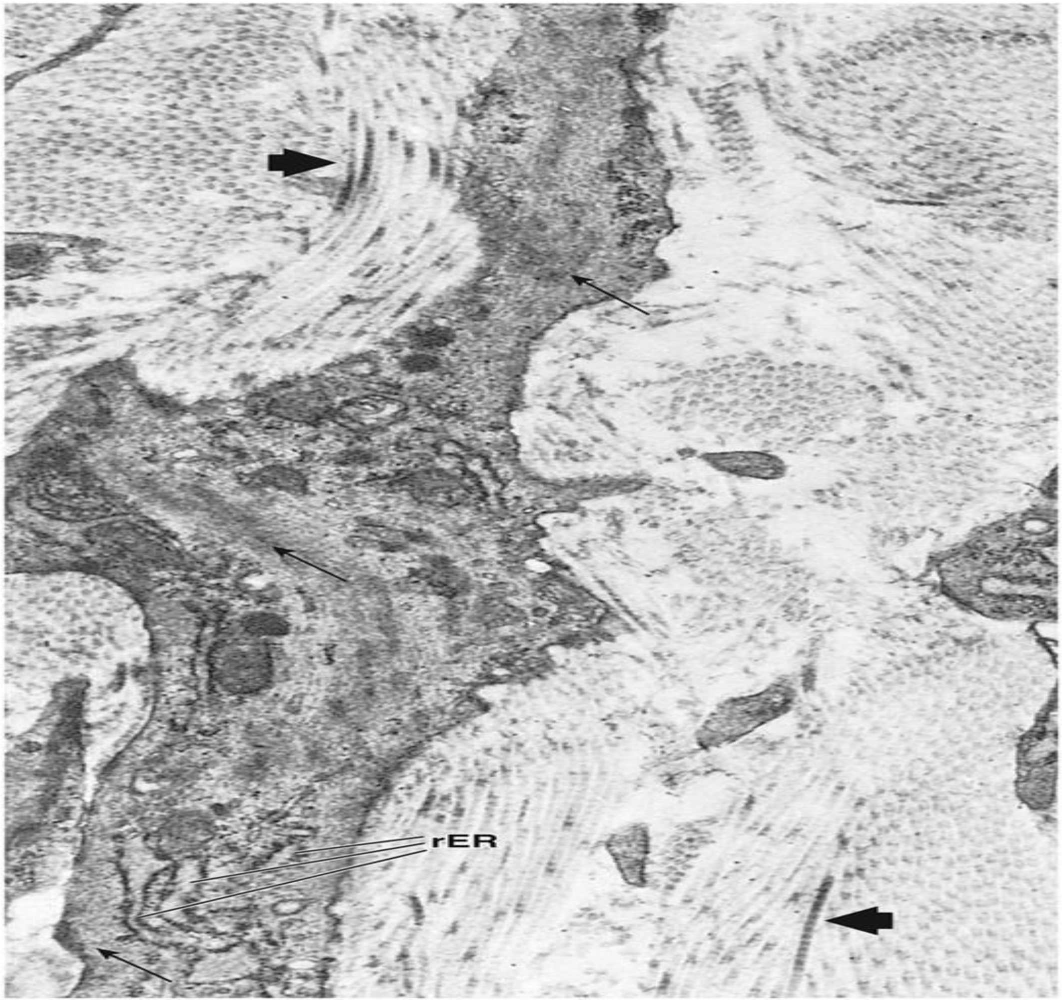


- In routine H&E preparations, only the nucleus is visible.
  - Appears as an **elongated or disc-like structure**
  - The thin, pale-staining, flattened processes that form the bulk of the cytoplasm are usually not visible, because they blend with the collagen fibers.



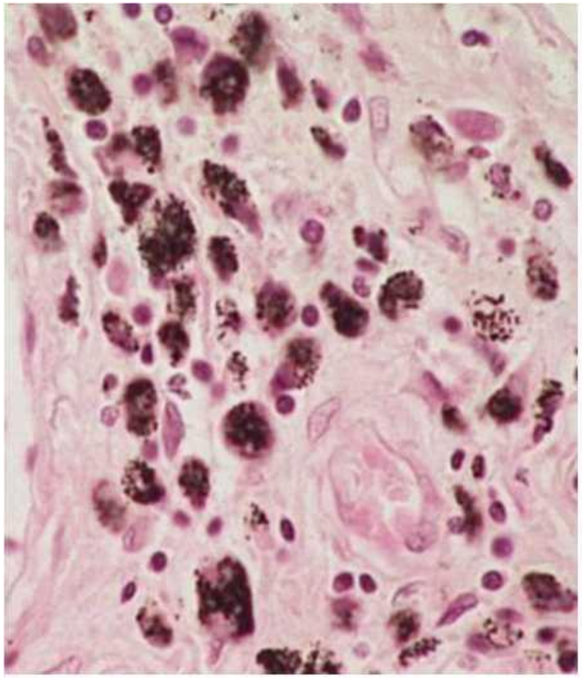
Myofibroblast

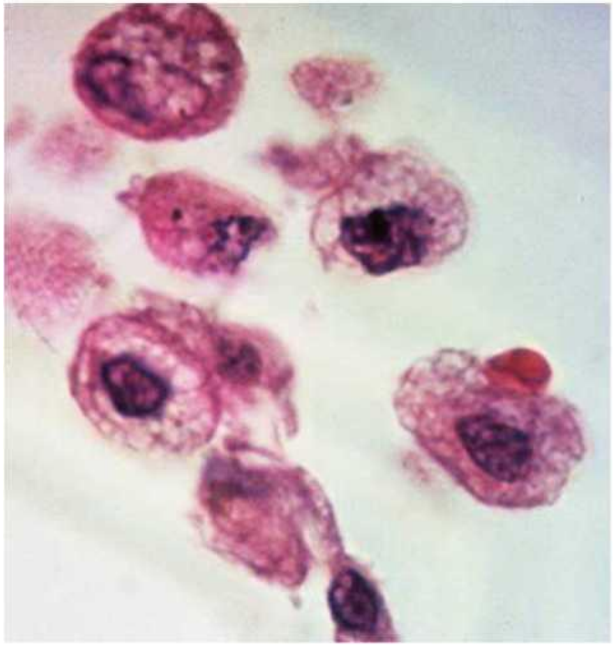
- Possesses some features of a fibroblast and characteristics of a smooth muscle cell
- Elongated, spindly and not easily identifiable on H&E
- Contains bundles of actin filaments with associated motor proteins
    - Moderate amount of rough endoplasmic reticulum
    - Lacks a surrounding basal lamina



Macrophages

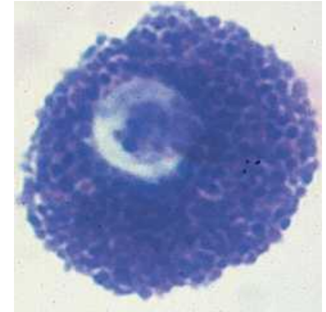
- Derived from monocytes
- Migrate to connective tissue and differentiate to macrophages
  - Liver – Kupffer cells
  - Brain – microglia
  - Bone – osteoclasts
  - Connective tissue – Histiocytes
  - Skin- Langerhans cells
- **Indented** or **kidney-shaped nucleus,** lysosomes are abundant in the cytoplasm
- Irregular cell membrane / cytoplasmic extensions (pseudopodia)
- Phagocytic; produce cytokines
- Antigen presenting cells; multinuclear giant cells

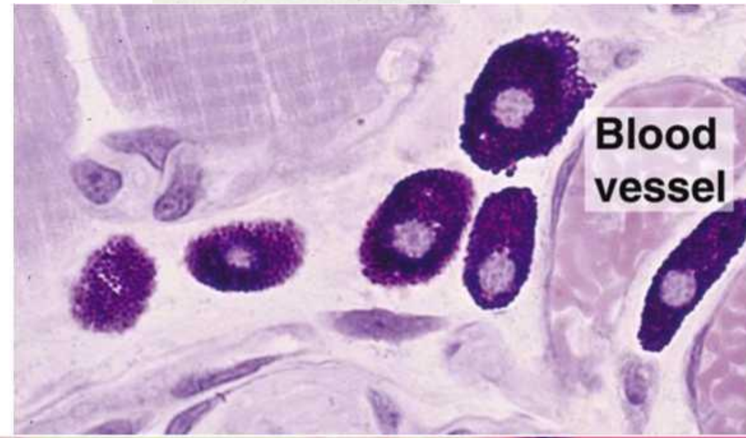




Mast Cells

- Originate in the bone marrow
- Migrate to connective tissue or lamina propria of mucosae
  - Proliferate and accumulate cytoplasmic granules in connective tissue
    - **Granules are large and intensely basophilic**
    - Requires special stains to visualize
  - Secretory products contained within the granules include **histamine**, **heparin**, **proteases**, and eosinophil and neutrophil chemotactic factors
- Possess receptors for **IgE antibodies** on their surface (allergic reactions)
- Exhibits **metachromasia**
  - Granules stain a different color from the color of the dye initially used to stain
    - Eg toluidine blue (a blue dye) stains mast cells purple-red instead of blue
    - This phenomenon is determined by a change in the electronic structure of the dye molecule after binding to the granular material
- In addition, mast cell granules are **PAS positive** because of their glycoprotein nature





Role of Connective Tissue Cells in Anaphylactic Shock

- **Mast Cells**
  - **Function in Anaphylaxis:**
    - **Antigen Recognition**: When an allergen binds to **IgE antibodies** (previously sensitized) on the surface of mast cells, it triggers rapid degranulation.
    - **Histamine Release**: Mast cells release **histamine**, a potent vasoactive mediator
- **Basophils**
  - **Function in Anaphylaxis:**
    - **IgG mediated,** results in the release of **platelet-activating factor (PAF)**
    - **Increased vascular permeability**
- **Eosinophils**
  - Eosinophils contribute to tissue damage and sustain the allergic response.
- **Fibroblasts**
  - It produces extracellular matrix proteins that help maintain tissue integrity. Chronic allergic reactions can stimulate fibroblast activation, leading to **tissue remodeling**.

Extracellular Matrix of Connective Tissue

- **The extracellular matrix (ECM)** is a complex and intricate structural network that surrounds and supports cells within the connective tissue.
- It contains a variety of **fibers** such as **collagen** and **elastic** fibers that are formed from different types of structural proteins.
- In addition, the ECM contains **Ground substance**
  - **Proteoglycans, glycosaminoglycans and multi-adhesive glycoproteins**

- Functions:
  1. Mechanical support
  2. Structural support
  3. Influence the extracellular communication
  4. Tensile strength
  5. Act as biochemical barrier and regulate the metabolic functions of the cells surrounded by the matrix

Extracellular Matrix of Connective Tissue: Ground Substance

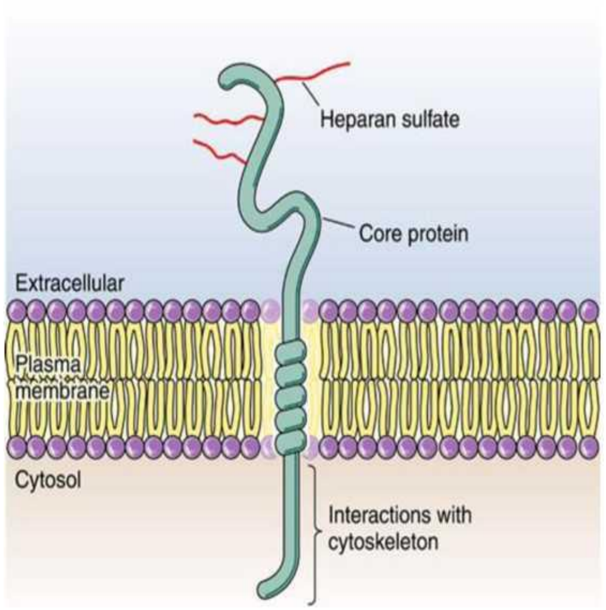
Ground Substance

- Found between the cells and fibers – space filler
- Clear, viscous substance with a slippery feel and high-water content
- Can be fluid, semi-fluid, gelatin-like or calcified – depending on the tissue
- Supports, bind, stores water, gives the tissue tensile strength and provides a medium for transport
- Contains numerous proteins and polysaccharides
    1. Glycosaminoglycans (GAGS)
    2. Proteoglycans (combination of GAGS with a protein core)
    3. Multi-adhesive glycoproteins

Components of the Ground Substance

Glycosaminoglycans (GAGs)

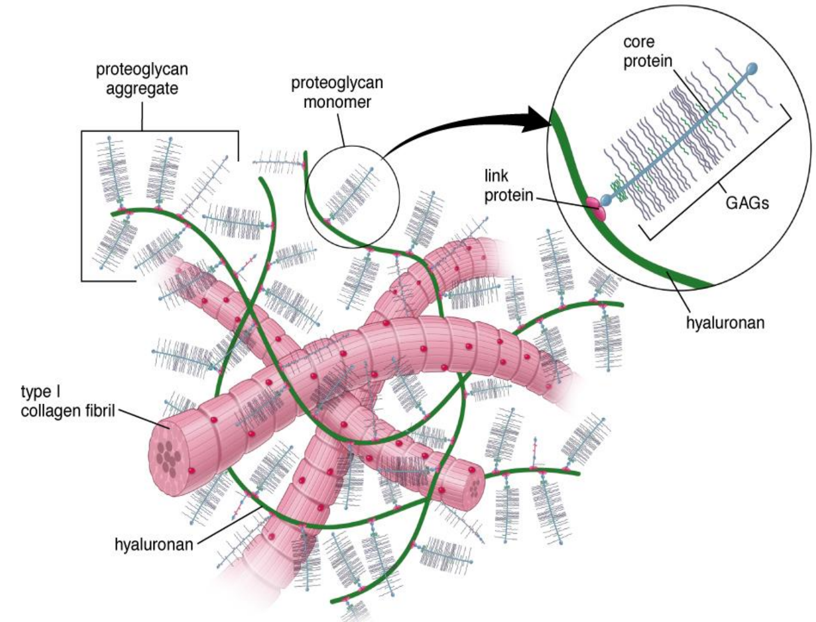
- Most abundant polysaccharide in ground substance
  - Long chain, unbranched
- Highly negatively charged
  - Possess carboxyl and sulfate groups on sugar molecules
  - Allows it to attract water, forming a hydrated gel
  - Stains well with basic dyes
- Permits rapid diffusion of water-soluble molecules
- E.g.
  - Hyaluronan
    - Extremely long and rigid
    - Can hold large volumes of water
    - Only GAG that is **not sulfated**, and **not modified post translation**
    - Abundant in **cartilage**
      - Acts as a shock absorber
  - Chondroitin-4-Sulphate
  - Chondroitin-6-Sulphate
  - Heparan sulphate
  - Keratan sulphate
- **GAGs are responsible for the physical properties of ground substance.**

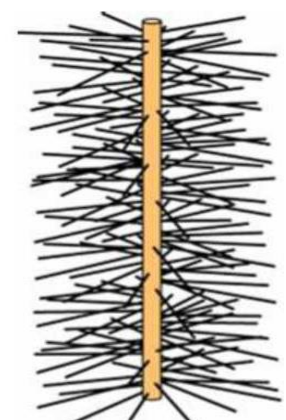


Components of the Ground Substance

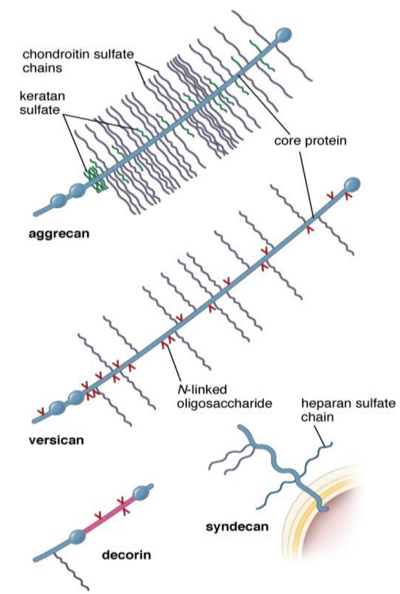
Proteoglycans (PGs)

- Composed of GAGs covalently attached to core proteins.
- GAGs extend perpendicularly from the core protein in a brush-like manner
  - Core protein has sites for multiple GAG attachments
    - Can have a single GAG attached (eg versican) or multiple different GAGS (syndecan or aggrecan)
      - Aggrecan contains both chondroitin sulfate and keratan sulfate
  - GAGs attachment can vary from 1 (as in the case of decorin) to hundreds (as in the case of aggrecan)
- Functions to link cells to ECM





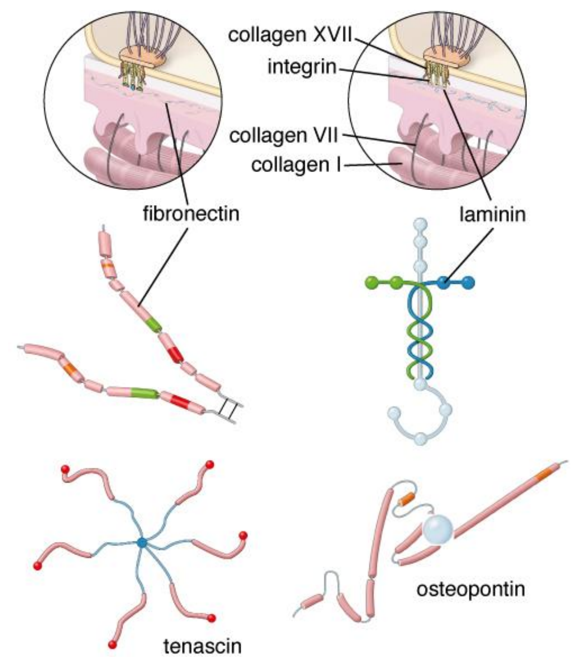
Bottle-brush like appearance



Components of the Ground Substance

Multi-adhesive Glycoproteins (MAGs)

- Have binding sites for ECM proteins such as collagens, GAGs and proteoglycans
- Functions
  - Stabilize the ECM and link it to the surface of connective tissue cells
  - Regulate cell movement and migration
  - Regulate cell differentiation and proliferation
- Examples:
  - Fibronectin
    - Most abundant glycoprotein
    - Aids in cell to ECM attachment
  - Laminin
    - Present in basal lamina
  - Osteoponin
    - Present in bone tissue
    - Helps attach osteoclasts to the bone surface
  - Tenascin
    - Important in wound healing
    - Most important during embryogenesis



Summary

	GAGS	PGs	MAGs
Structure	Long chain, unbranched polysaccharides	GAGs covalently attached to core proteins.	Protein Core Carbohydrate Chains
Function	Physical properties of ground substance.	link cells to ECM	•Stabilize the ECM and link it to the surface of connective tissue cells •Regulate cell movement and migration •Regulate cell differentiation and proliferation
Examples	Chondroitin-4-Sulphate Chondroitin-6-Sulphate Heparan sulphate Keratan sulphate Hyaluronan	Versican Syndecan	Fibronectin Laminin Osteoponin Tenascin

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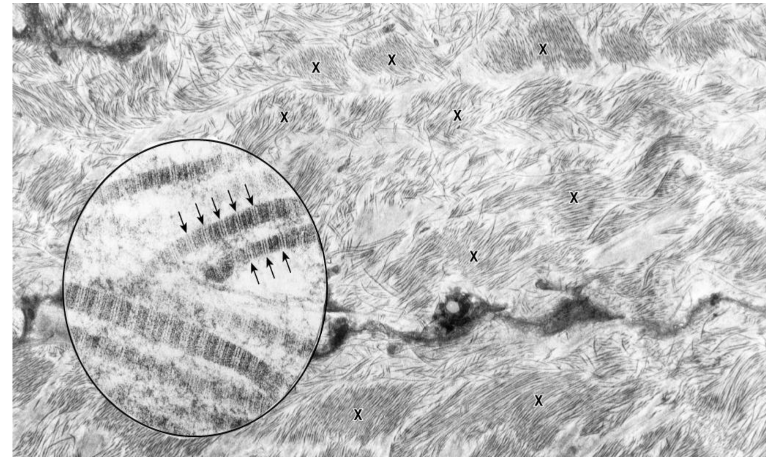
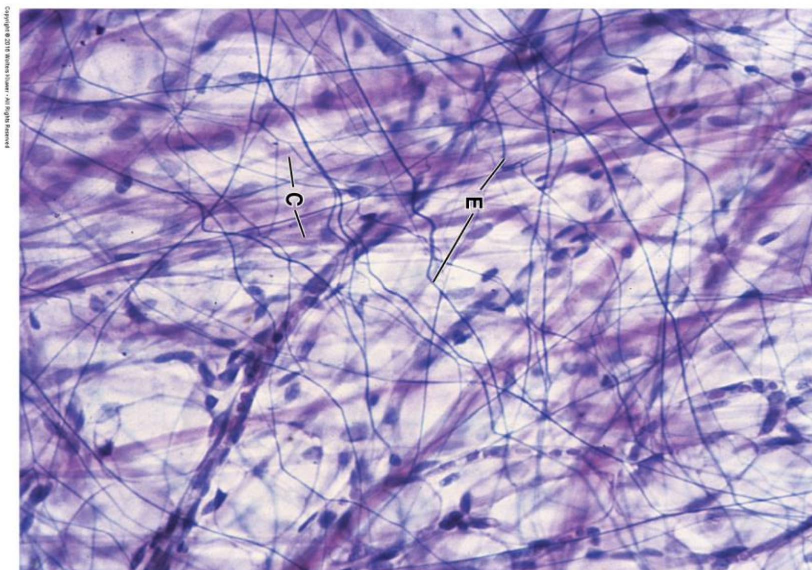
Fibers of Connective Tissue: Collagen

- Most abundant type of fibers in connective tissue
- Long polymer
    - Unbranched, thick and wavy (under light microscope)
    - Appears eosinophilic with hematoxylin & eosin dyes
- Produced by fibroblasts
    - Has various names in different tissues
        - Chondrocytes (cartilage)
        - Osteoblasts (bone)
        - Pericytes (blood vessels)
- Many different types with **type I** being the most abundant
- Different classes:
    - **Fibrillar collagen**
        - Provides mechanical strength
            - E.g. tendons, bones – type I
        - Provides resistance to pressure
            - E.g. cartilage – type II
        - Provides scaffolding for cells
            - E.g. reticular – type III
    - **Basement membrane collagen**
        - Collagens type IV collagen, VI, VII

Collagen fibrils - EM



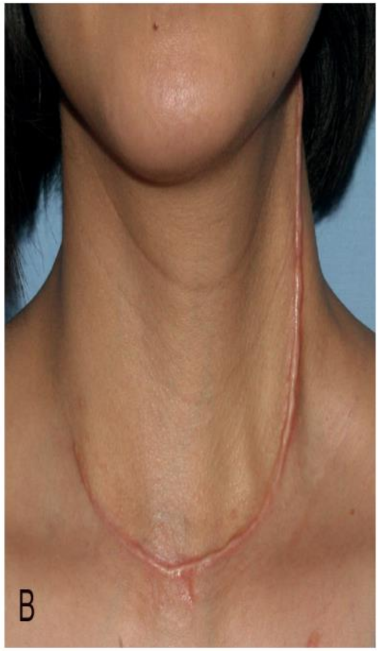


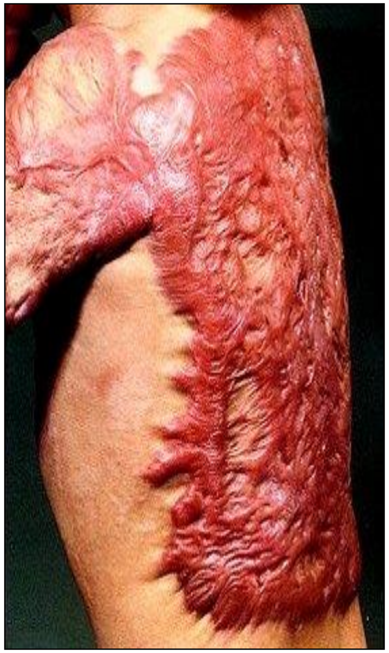
Collagen fibrils – Light microscopy

Fibers of Connective Tissue: Collagen

- **Hypertrophic scar→** Scar when more raised than normal, but within original wound boundary

- **Keloid scar→ increased collagen production**
    - scar extends into surrounding tissue (beyond original wound boundary)

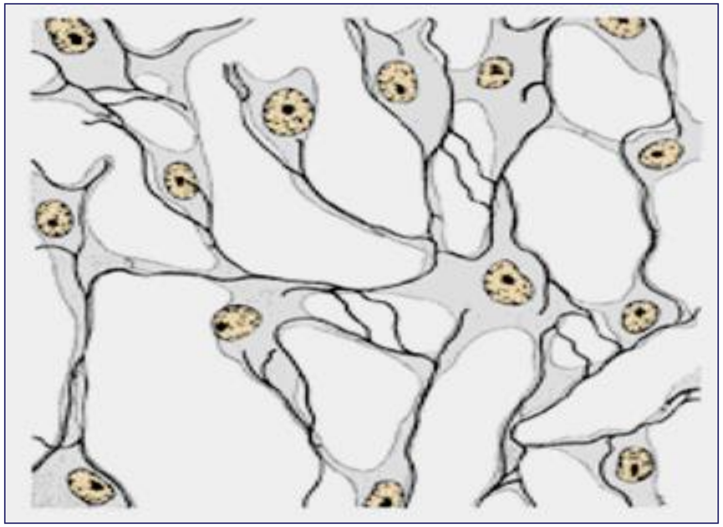


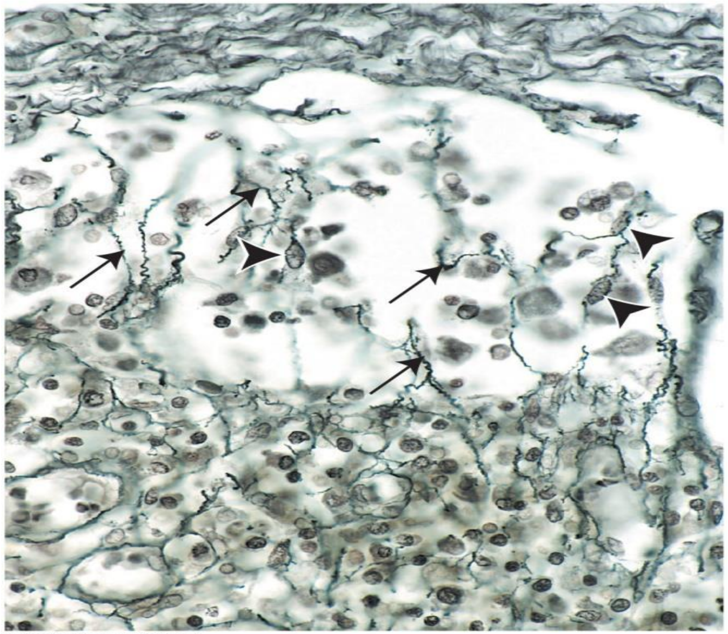


Principles of Wound Healing - ENT Secrets Sokoya, Mofiyinfolu, MD; Winkler, Andrew A., MD;

Fibers of Connective Tissue: Reticular

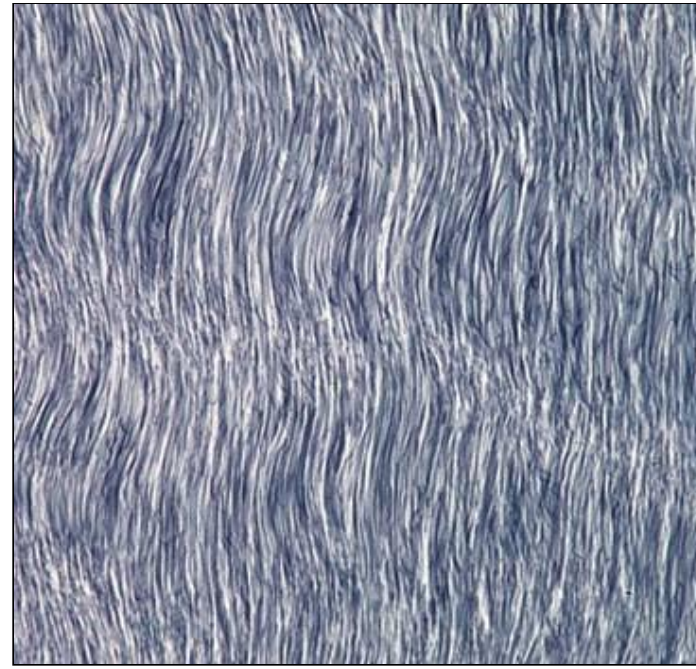
- Contain **type III collagen fibers**
- Short, thin, and highly branching in nature
  - Secreted by fibroblasts in most locations
- **Argyrophilic** – has a **high affinity for silver nitrate** stains and also stains well with **PAS**
- First type of collagen to be synthesized during wound healing (most abundant during early stages of healing)
- Forms basement membrane along with basal lamina
- Can also be secreted by
  1. Reticular cells(hemopoietic and lymphatic tissues)
  2. Schwann cells (peripheral nerves)
  3. Smooth muscle cells(tunica media of blood vessels; and muscularis layer of the alimentary canal wall)
- **Function:**
  - Forms a **supporting framework for the cellular constituents** of various tissues and organs including
    - Hemopoietic and lymphatic tissue→ bone marrow, lymph nodes, spleen (*not Thymus)
    - Liver
    - Endocrine tissue
    - Nerve tissue

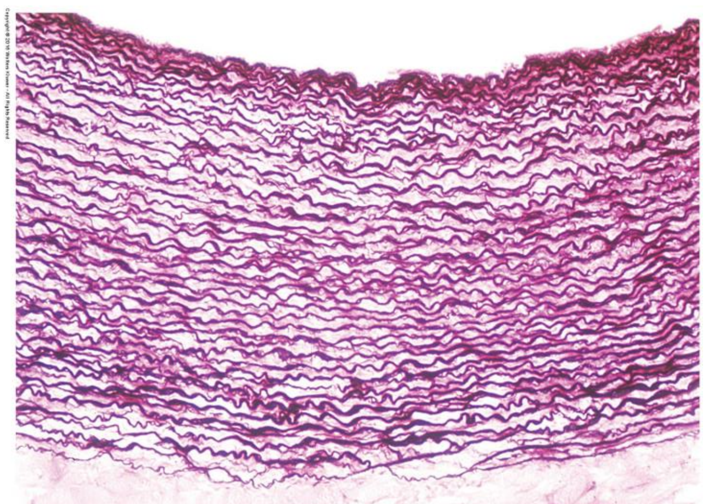




Fibers of Connective Tissue: Elastic

- Normally **long, thin and branching**
  - Forms network
- Elastic fibers allow tissues to respond to stretch and distension.
- Composed of cross-linked **elastin molecules** and a network of **fibrillin microfibrils** with **associated proteins (EMILIN-1, MAGP-1)**
- Found in large arteries, elastic cartilage, vocal ligament, bronchi, ligamentum flavum, suspensory ligament of the penis, skin etc.
- Special staining required - orcein, resorcin or Verhoeff's
- May also be produced by smooth muscle cells (blood vessels), endothelial cells (blood vessels) and chondrocytes (cartilage)





Elastic artery

Summary

	Collagen fibers	Elastic fibers	Reticular fibers
Description under LM	Unbranched, thick, and wavy	long, thin and branching	Short, thin and highly branching
Stain used	H & E - Trichrome	Elastic stains	PAS, Silver stains
Function	Provides Strength and resilience to pulling and stretch	Allow tissues to respond to stretch and distension.	Supporting framework for the cellular constituents
Location	• Type I: bone, tendons • Type II: Cartilage • Type III: Reticular fibers • Type IV,VI, VII: basement membrane	• Large arteries • elastic cartilage • Vocal ligament • Bronchi • Ligamentum flavum, • Suspensory ligament of the penis • Skin	• Hemopoietic and lymphatic tissue (bone marrow, lymph nodes, spleen) • Liver • Endocrine tissue • Nerve tissue

Classification of Connective Tissue

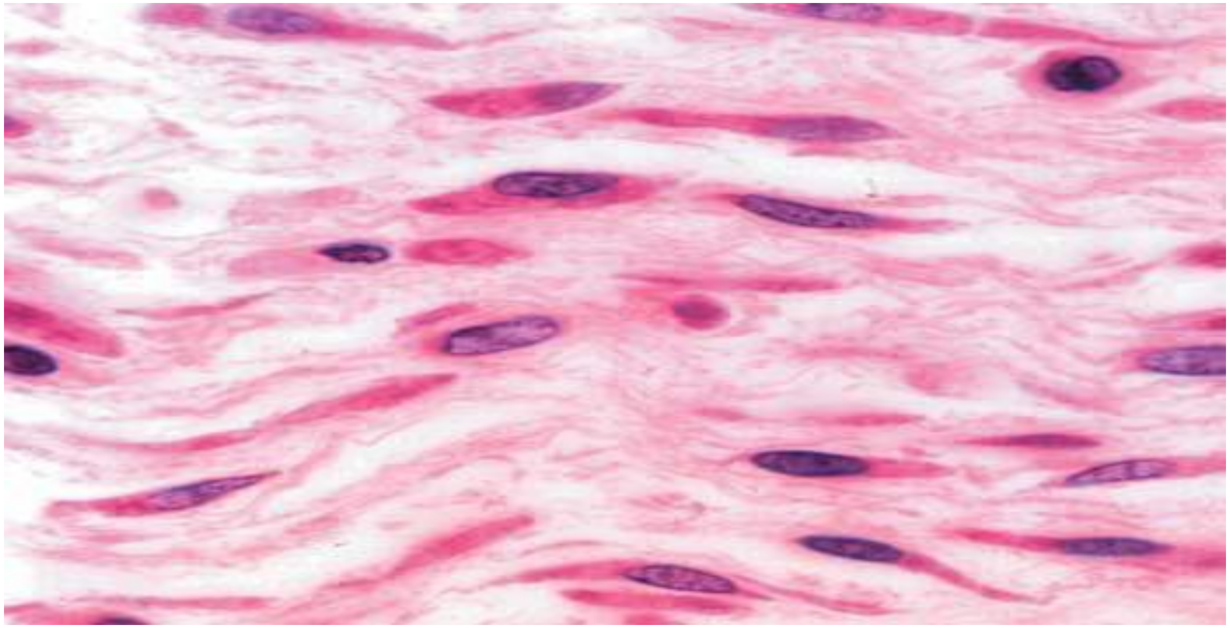
1. **Embryonic connective tissue**
   a) Mesenchyme
   b) Mucous

2. **Connective tissue proper**
   a) Loose
   b) Dense
      1. Regular
      2. Irregular

3. **Specialized connective tissue**
   a) Adipose
   b) Blood
   c) Bone
   d) Cartilage

Embryonic Connective Tissue: Mesenchyme

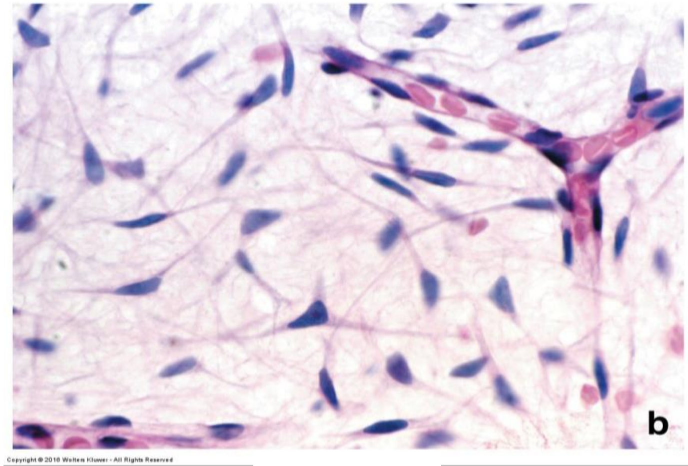
- Gives rise to almost all connective tissue types
- Primarily found in the **embryo**
- It contains small, spindle-shaped cells
- These cells have **processes** that extend to neighboring cells
  - **Gap junctions** are found here
- The extracellular space is occupied by viscous ground substance
- Collagen and reticular fibers are very fine and sparse.



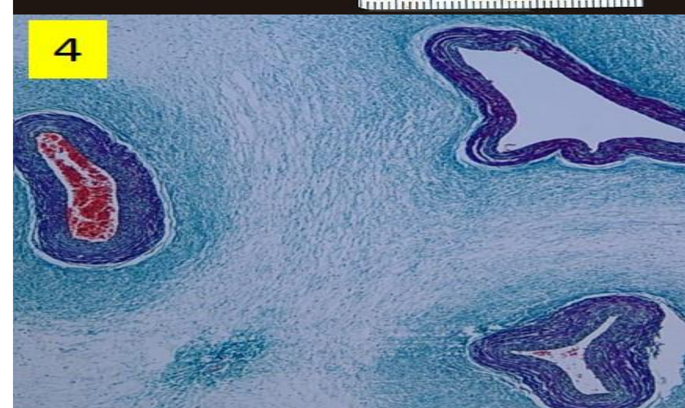
H&E x480

Embryonic Connective Tissue: Mucous

- Present in the **umbilical cord**
- Consists of gelatine-like ECM; composed mainly of hyaluronan.
- Its ground substance is frequently referred to as **Wharton's jelly.**
- The spindle-shaped cells are widely separated and appear much like fibroblasts (cytoplasmic processes are thin and difficult to visualize in routine H&E preparation).



H&E x480



Connective Tissue Proper: Loose (Areolar)

- Contains an **abundance of varying cells** and sparse, loosely arranged fibers
- Most cells are wandering cells
- **Abundant,** gel-like ground substance: it plays an important role in the diffusion of oxygen, nutrients, carbon dioxide, and metabolic waste.
- Rich blood supply
- Location: the lamina propria of mucous membranes (e.g. urinary, respiratory, and digestive tracks), surrounds blood vessels and glands

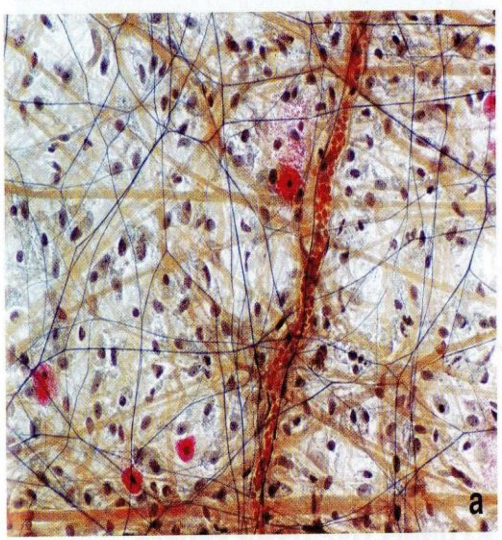
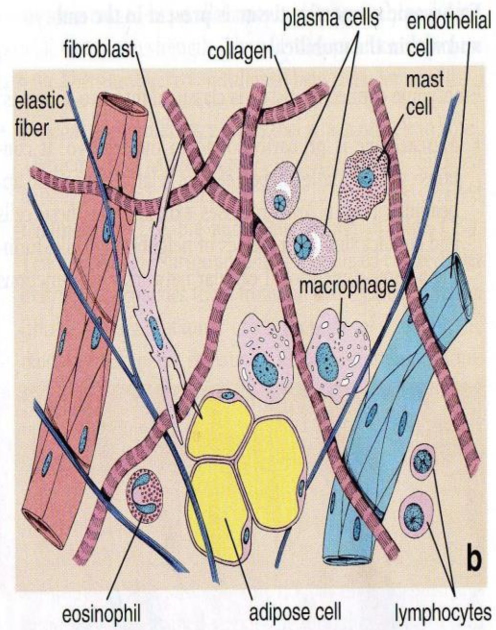


Fig A: Whole Mount Preparation of Mesentery



Diagrammatic representation of whole mount preparation

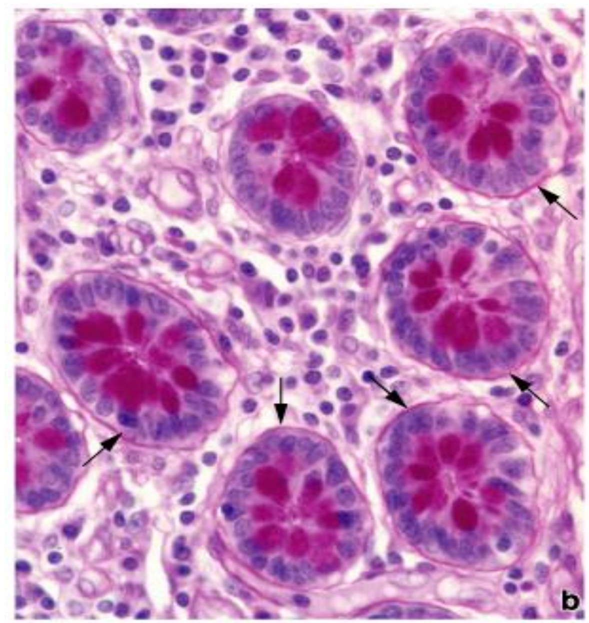
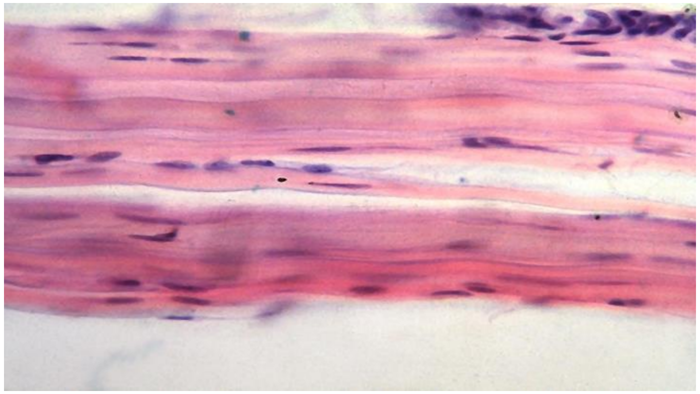
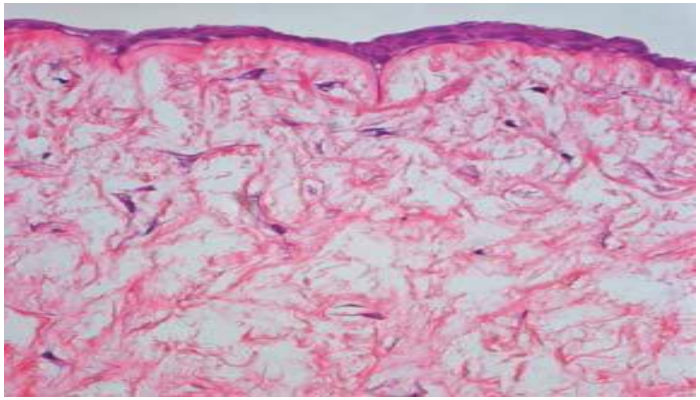


Figure 4.17b. Basement membrane: photomicrograph of intestinal glands of the colon in cross-section stained with the PAS method. *Arrows,* basement membrane. X550.

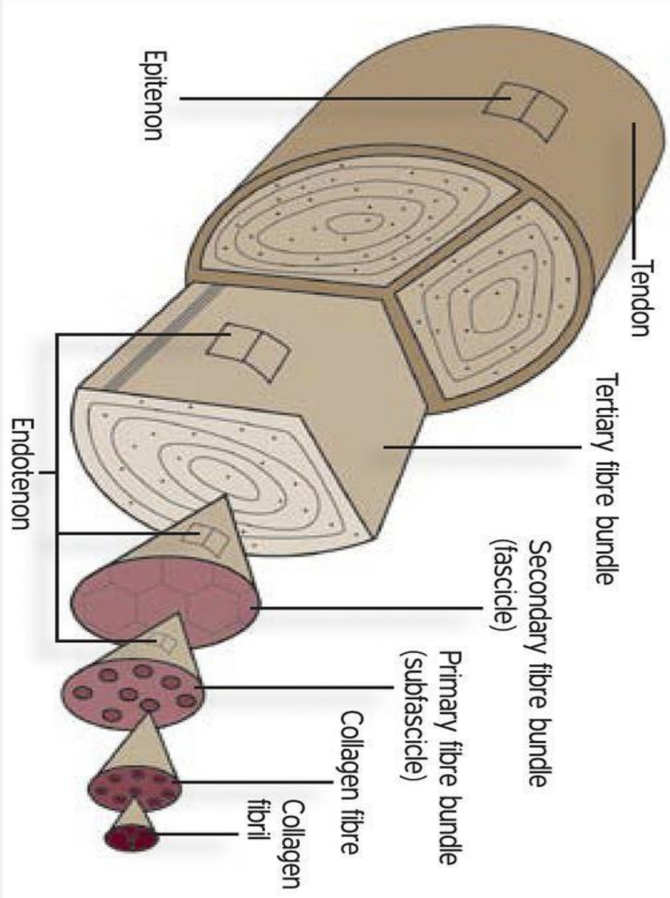
Connective Tissue Proper: Dense

- Contains abundant fibers (typically collagen type I) and fewer cells (mainly fibroblasts)
- Little ground substance
- Further categorized based on the organization of the collagen fibers
  - **Dense Regular** – parallel rows of collagen fibers
    - Withstands unidirectional stress e.g., tendons, ligaments, aponeuroses
  - **Dense Irregular** – collagen fibers haphazardly arranged
    - Withstands tension from all direction e.g., dermis of the skin, periosteum, organ capsules

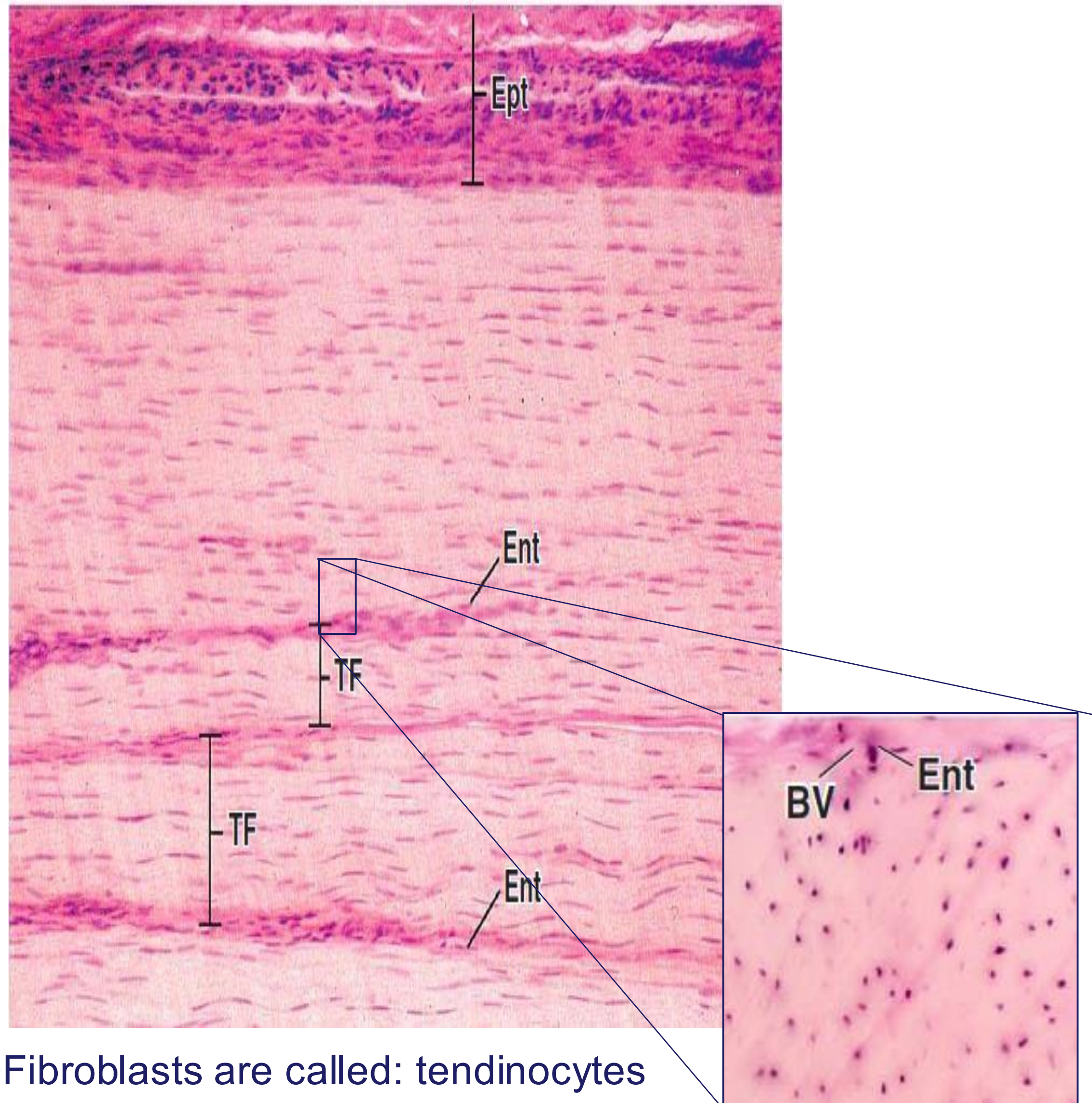




Dense Regular Connective Tissue: Tendon



- The substance of the tendon is surrounded by a thin connective tissue capsule, **the epitendineum**,
- The tendon is subdivided into fascicles by **endotendineum**

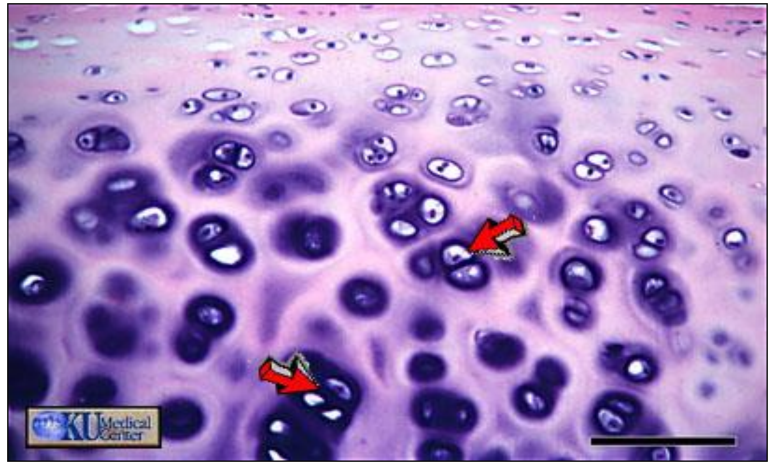


•Fibroblasts are called: tendinocytes
•Collagen fibers→ type 1 (imparts tensile strength)

Specialized Connective Tissue

1. Cartilage
2. Bone
3. Blood

1



2

Sectional view of an osteon (haversian system) of femur (thigh bone)

3

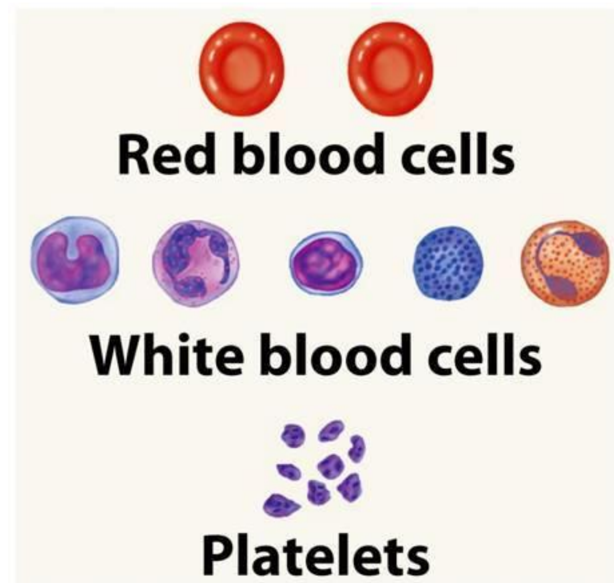


Table 4-4 figure 22 Principles of Anatomy and Physiology, 11/e
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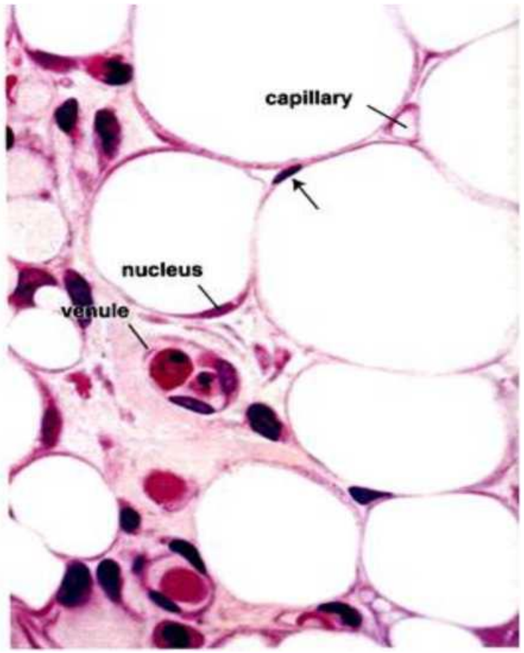
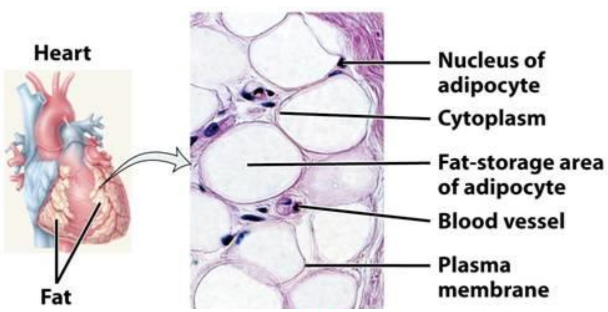
Adipose Connective Tissue

Features	White	Brown
Cell size	Large	Small
Nucleus location	Peripheral	Central
Lipid droplet	Unilocular = one large central	Multilocular= Multiple
Color	White/ yellow [carotenoid]	Brown [cytochrome oxidase in mitochondria]
Location	Widely distributed, affected by age and gender	Postnatal life and back of the neck
Function	Energy storage, insulation, cushioning and hormones production	Heat production (thermogenesis)

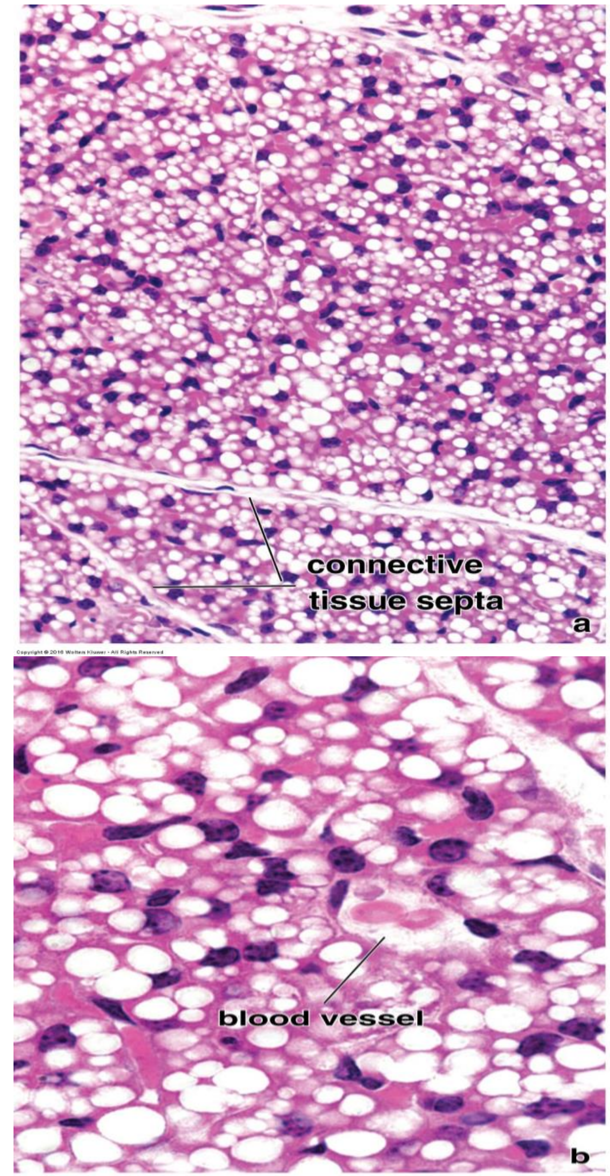
Adipose tissue

White adipose tissue

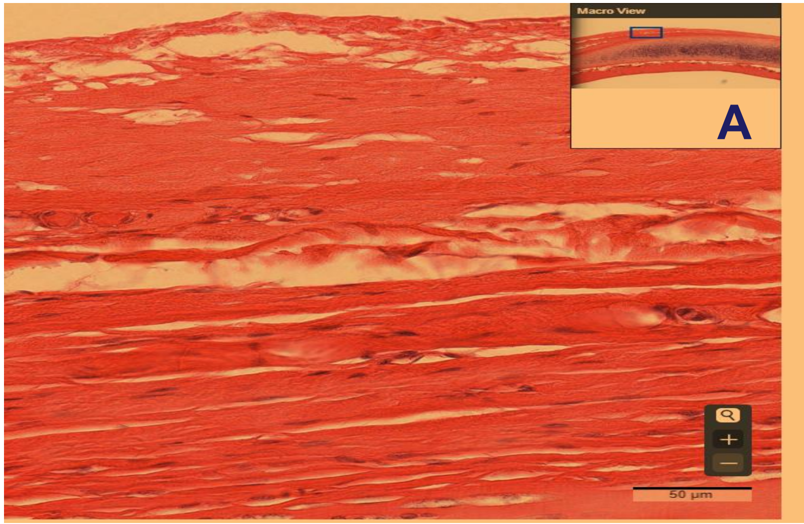




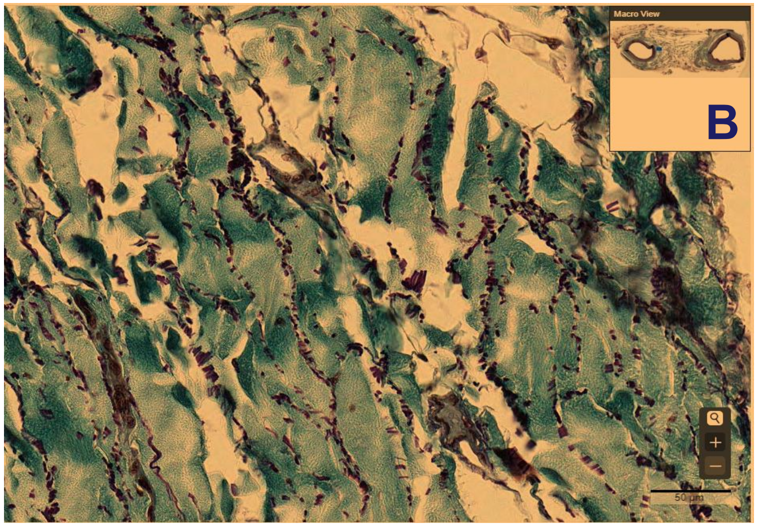
Brown adipose tissue



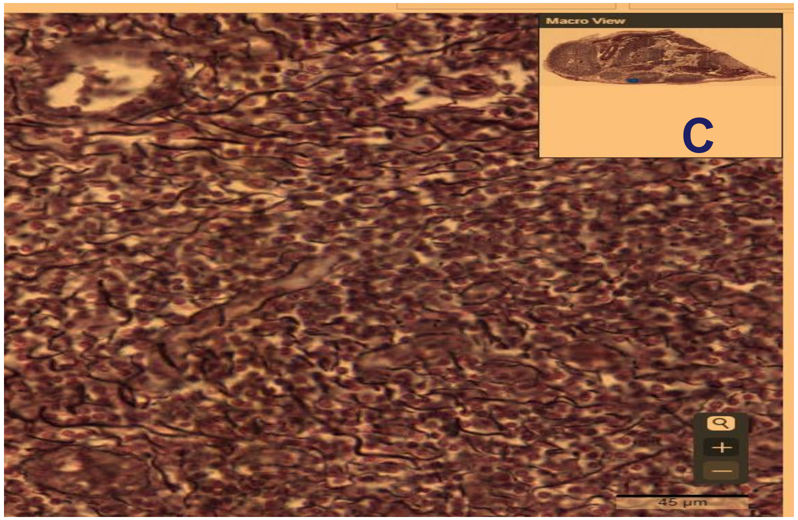
The following stains are commonly used to identify connective tissues for observation with light microscopy



Hematoxylin & Eosin stain – most common



Trichrome stain – differentiates between nucleus, cytoplasm and connective tissue fibers

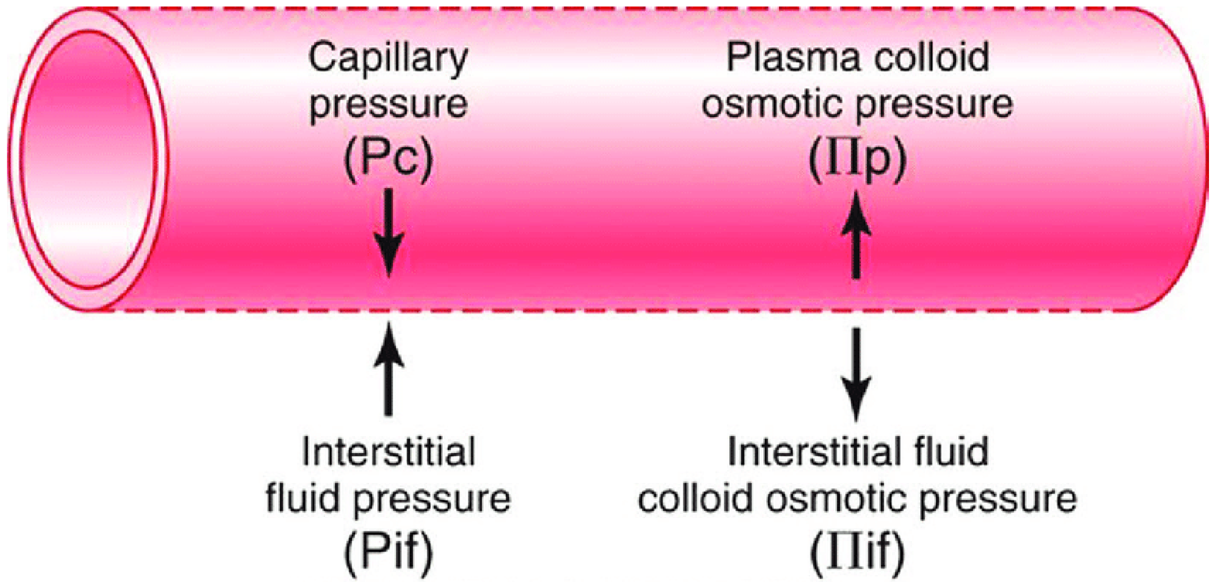


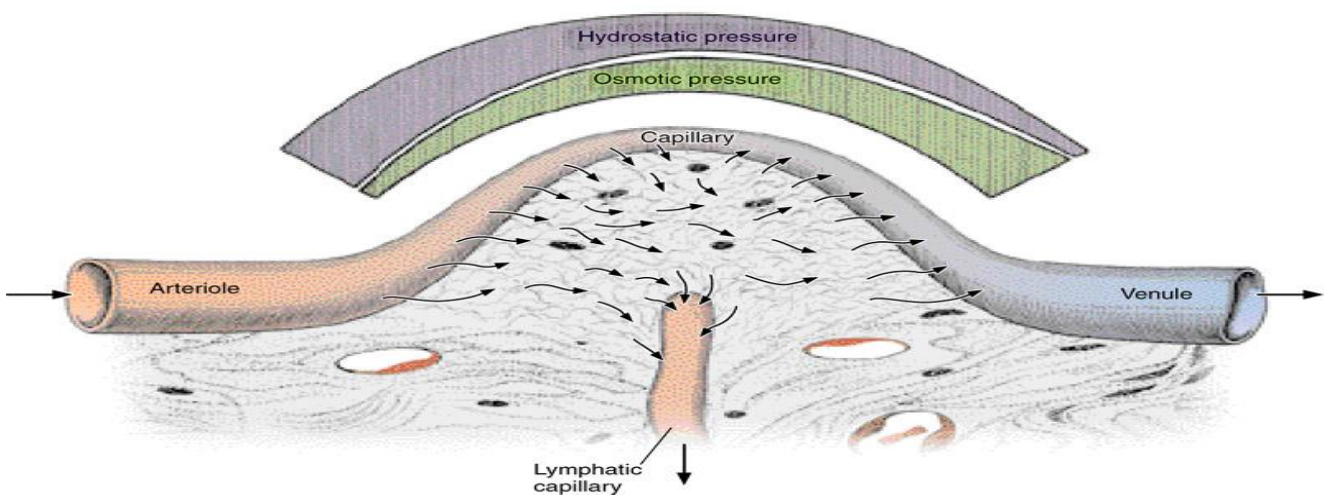
Silver stain – used to stain reticular fibers stains them black

Elastic stain – stains elastic fibers Eg Orcein, Verhoeff's

Formation of Interstitial Fluid, Lymph, and Edema

- **Formation of Interstitial Fluid**
  - Interstitial fluid is the fluid that surrounds the cells of tissues, providing them with nutrients and oxygen while removing waste products. It forms due to the balance between hydrostatic and osmotic pressures in the capillary microcirculation.
  - **Capillary Hydrostatic Pressure (CHP)**: This is the pressure exerted by blood inside the capillaries, pushing water and small solutes **out** of the **capillaries** into the interstitial space.
  - **Capillary Oncotic Pressure (COP)**: Plasma proteins (mainly albumin) generate an osmotic force that pulls water
  - **Interstitial Hydrostatic Pressure (IHP)**: This is the pressure exerted **back into the capillaries** by interstitial fluid, which can push fluid **back into the capillaries**.
  - **Interstitial Oncotic Pressure (IOP)**: Proteins in the interstitial space attract fluid **out of the capillaries.**
- A delicate balance between these forces, known as **Starling's forces**, determines the movement of fluid in and out of capillaries. When filtration exceeds reabsorption, excess fluid remains in the interstitial space, forming interstitial fluid.


Capillary pressure (Pc)
Plasma colloid osmotic pressure (Πp)
Interstitial fluid pressure (Pif)
Interstitial fluid colloid osmotic pressure (Πif)


Hydrostatic pressure
Osmotic pressure
Capillary
Arteriole
Venule
Lymphatic capillary

Formation of Lymph

Lymph formation is a consequence of interstitial fluid drainage via the lymphatic system.

- As extracellular fluid leaks out of the blood capillary walls due to forces exerted by the heart or osmotic pressure, interstitial fluid accumulates
- This fluid is collected by small lymphatic capillaries with other substances to form lymph

- **Role of the Lymphatic System**: Lymphatic capillaries absorb excess interstitial fluid, along with proteins and cellular debris, preventing fluid accumulation in tissues.
- **One-Way Flow**: Lymphatic vessels contain valves that ensure unidirectional flow toward larger lymphatic ducts and ultimately return the fluid to the bloodstream via the subclavian veins.
- **Composition of Lymph**: It consists of water, proteins, immune cells (like lymphocytes), lipids (from intestinal absorption), and cellular debris.
    - Similar to blood plasma; watery, transparent and yellowish in appearance

Development of Edema

- Edema refers to the excessive accumulation of interstitial fluid, leading to tissue swelling. It results from an imbalance in Starling's forces or lymphatic dysfunction.
- **Causes of Edema**
  1. **Increased Capillary Hydrostatic Pressure**:
     1. Hypertension (high blood pressure)
     2. Congestive heart failure (leading to venous congestion)
     3. Deep vein thrombosis (DVT)
  2. **Decreased Capillary Oncotic Pressure**:
     1. Hypoalbuminemia (e.g., in liver disease, nephrotic syndrome, or malnutrition)
     2. Severe burns (loss of plasma proteins)
  3. **Increased Capillary Permeability**:
     1. Inflammatory responses (infection, trauma, allergic reactions)
     2. Sepsis or toxins causing endothelial damage
  4. **Lymphatic Obstruction (Lymphedema)**:
     1. Tumors compressing lymphatic vessels
     2. Surgical removal of lymph nodes (e.g., after mastectomy)
     3. Parasitic infections (e.g., filariasis)
  5. **Sodium and Water Retention**:
     1. Kidney diseases (e.g., chronic kidney disease, nephrotic syndrome)
     2. Heart failure leading to increased aldosterone secretion

Summary

- **Interstitial fluid** forms due to the balance of hydrostatic and oncotic pressures in capillaries.
- **Lymph** is interstitial fluid drained by the lymphatic system to prevent accumulation.
- **Edema** develops when fluid accumulation in the interstitial space exceeds lymphatic drainage, often due to increased capillary pressure, reduced oncotic pressure, increased permeability, or lymphatic obstruction.